Split-Brain Scenarios

when two nodes both think they're the leader

Phase 4 videos on YouTube

In one line: split-brain happens when a cluster splits and both sides act as leader, accepting conflicting writes; prevent it with majority quorums, an odd number of voters (or a witness), and fencing that stops the old leader, because 'it stopped answering' never proves it stopped working.

1 Two primaries

Your database primary stops answering health checks. The failover system promotes a replica. But the old primary wasn't dead; it was cut off from the monitor, and app servers next to it are still writing to it. Now there are **two primaries** accepting writes, and their data diverges. That's **split-brain**, and it's one of the most damaging failures in distributed systems, because it silently corrupts data instead of just causing downtime.



Figure 1. The monitor can't tell 'dead' from 'unreachable' (lessons 3 and 4). Promoting without stopping the old primary creates two leaders.

2 Why it happens

- Failover decided by one observer that is itself partitioned.
- Two-node clusters: after a split, each side is exactly half, and a naive rule lets both win.
- A leader paused by GC or a frozen VM, which wakes up still believing it leads (lesson 8).
- Manual failover during an incident, without first making sure the old primary is down.

WORKED EXAMPLE: POSTGRES WITH A HOMEMADE FAILOVER SCRIPT

A Postgres cluster has a primary and one replica, with a simple script that promotes the replica if the primary fails three health checks. A network glitch hides the primary from the script for 20 seconds. The replica is promoted, DNS is switched, but some app servers keep their old connections to the original primary. For ten minutes, orders are written to both. Cleaning up means comparing two databases row by row. The fix: a tool like Patroni that keeps the leader key in etcd with a lease, a primary that **demotes itself** when it can't renew the key, and fencing (revoke its network access or kill it) before promotion.

3 Defences

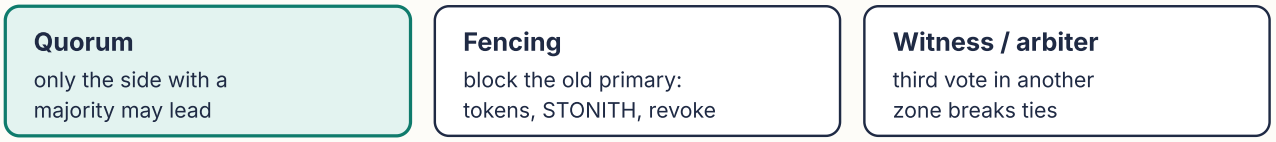
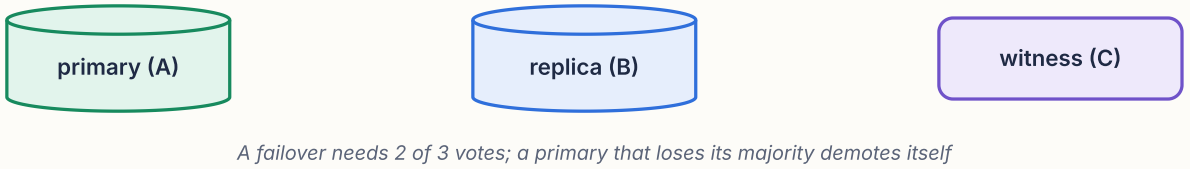


Figure 2. Majority, fencing and a tie-breaking witness: use them together.

Quorum: a node may act as leader only while it can reach a majority of voters; the minority side stops. **Fencing:** make it impossible for the old leader to do harm: fencing tokens checked by storage, revoking its credentials or network access, or power-cycling it (STONITH, 'shoot the other node in the head'). **Witness:** with only two data nodes, add a lightweight third voter in another zone so a majority always exists.



A failover needs 2 of 3 votes; a primary that loses its majority demotes itself

Figure 3. Two data nodes plus a witness: three voters, so one side of any split has a majority.

COMMON MISTAKES

- Failing over on one observer's opinion.
- Two-node clusters with automatic failover and no witness.
- Promoting a replica without fencing the old primary.
- Relying on DNS changes to move all clients instantly.

4 Key terms

Split-brain: two nodes act as leader at once

STONITH: forcibly power off the other node

Self-demotion: a leader steps down when it loses its lease or quorum

Fencing: stopping the old leader from acting

Witness / arbiter: voter without data that breaks ties

5 Interview questions

Q1. What is split-brain? BASIC

A state where two parts of a cluster each believe they're the leader and accept writes independently, so the data diverges.

Q2. How do quorums prevent it? BASIC

Only a side with a majority of voters may elect or keep a leader; two disjoint majorities can't exist, so at most one side leads.

Q3. You have two database nodes in two zones. How do you make automatic failover safe? SCENARIO

Add a witness in a third zone so there are three voters, keep leadership in a consensus store with a lease, and fence the old primary before promoting.

Q4. Split-brain already happened for ten minutes. What now? SCENARIO

Stop writes to one side, choose a survivor, find writes made only on the other side, reconcile them (replay or manual review), then rebuild the losing node as a replica.

Q5. Why isn't 'the primary stopped answering health checks' enough to promote a replica? TRICKY

The primary may only be unreachable from the monitor while still serving clients. Without fencing, promotion creates a second primary.

Q6. What's the role of fencing tokens here? TRICKY

Storage rejects writes with an older token, so even if a deposed leader keeps running and writing, its writes are refused.

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